

SUMY, Ukraine

WMO: 332750

Lat: 50.8589N

Lon: 34.7492E

Elev: 181

StdP: 99.17

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.6	-17.7	-23.6	0.5	-20.1	-20.7	0.6	-17.2	10.3	-1.8	9.3	-3.0	2.4	30	0.559

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	31.2	20.2	29.4	19.5	27.7	18.8	21.5	28.1	20.6	27.1	19.8	26.0	3.7	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	14.4	24.4	18.5	13.6	23.4	17.7	13.0	22.5	63.3	28.0	60.3	27.2	57.5	26.1	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.1	8.2	7.4	-24.1	33.6	3.0	2.1	-26.2	35.1	-27.9	36.3	-29.6	37.5	-31.7	39.1
			-24.3	23.1	2.9	1.4	-26.4	24.1	-28.1	24.9	-29.8	25.7	-31.9	26.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	-5.6	-4.7	0.4	8.9	15.1	18.6	20.6	19.5	13.9	7.3	1.4	-3.5
	DBStd	10.52	5.88	6.09	4.64	4.32	4.49	3.62	3.15	3.64	4.09	4.45	4.80	5.76
	HDD10.0	2069	484	412	297	71	9	0	0	0	11	107	259	418
	HDD18.3	4101	743	645	556	284	120	42	13	29	143	343	508	677
	CDD10.0	1238	0	0	1	38	167	258	329	295	127	23	1	0
	CDD18.3	228	0	0	0	1	19	49	84	66	9	0	0	0
	CDH23.3	2093	0	0	0	7	194	399	723	671	97	2	0	0
	CDH26.7	613	0	0	0	1	40	89	224	241	18	0	0	0
Wind	WSAvg	3.6	4.1	4.3	4.2	3.8	3.5	3.3	2.8	2.9	3.1	3.5	4.0	4.1
Precipitation	PrecAvg	603	44	32	36	39	56	66	75	60	47	43	52	49
	PrecMax	877	129	71	80	116	156	140	178	167	107	158	145	123
	PrecMin	388	1	3	6	5	6	13	13	4	5	1	10	8
	PrecStd	124	29	18	20	25	35	30	40	34	30	36	32	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.5	6.8	15.2	24.0	30.0	30.9	33.5	35.2	28.8	22.0	13.9	8.7
		MCWB	4.8	5.0	8.7	13.4	18.4	19.7	21.1	19.8	17.3	14.7	11.4	7.9
	2%	DB	3.5	4.4	11.5	21.3	27.6	29.1	30.9	31.6	26.3	18.8	11.1	6.2
		MCWB	2.7	3.5	6.9	12.4	16.9	19.3	20.6	20.2	16.1	13.8	9.5	5.6
	5%	DB	2.2	2.8	8.7	18.7	25.6	27.4	29.2	29.3	23.5	15.8	9.2	4.5
		MCWB	1.6	2.0	5.4	11.3	16.6	18.9	20.2	19.4	15.6	12.4	8.1	4.0
	10%	DB	1.2	1.8	6.5	16.5	23.2	25.8	27.4	27.0	21.0	13.5	7.6	2.7
		MCWB	0.7	1.1	4.1	10.1	15.5	18.1	19.6	18.4	14.6	10.9	6.7	2.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.1	5.2	9.4	15.0	19.5	21.8	22.9	22.5	18.6	16.3	11.9	8.0
		MCDB	5.4	6.2	14.0	21.6	27.0	27.5	30.2	29.5	24.5	19.6	13.4	8.7
	2%	WB	2.8	3.6	7.4	13.2	18.3	20.7	21.7	21.0	17.4	14.3	10.0	5.7
		MCDB	3.3	4.3	10.6	19.0	25.3	26.7	28.5	28.4	23.0	17.6	10.8	6.1
	5%	WB	1.7	2.1	5.7	12.1	17.2	19.8	21.0	20.1	16.5	12.7	8.2	4.1
		MCDB	2.1	2.6	8.2	17.0	23.6	25.6	27.3	27.8	21.6	15.3	9.0	4.5
	10%	WB	0.7	1.2	4.3	11.0	16.2	18.9	20.2	19.2	15.4	11.4	6.8	2.3
		MCDB	1.2	1.7	6.3	15.5	21.8	24.3	25.8	26.1	20.1	13.3	7.4	2.7
Mean Daily Temperature Range	5% DB	MDBR	4.7	5.5	6.5	9.4	10.5	9.7	10.1	11.1	9.8	7.5	4.5	4.3
		MCDBR	3.4	4.8	10.3	12.9	13.4	11.9	12.5	14.5	14.0	11.6	6.8	4.5
		MCWBR	3.3	4.2	6.4	6.2	5.4	4.9	4.5	5.0	6.2	6.8	5.4	4.4
	5% WB	MCDBR	3.3	4.5	9.2	10.9	11.7	10.5	11.1	12.9	11.8	10.0	5.8	4.4
		MCWBR	3.3	4.2	6.2	5.7	5.4	5.0	4.8	5.1	6.0	6.5	5.0	4.4
Clear-Sky Solar Irradiance	taub	0.307	0.339	0.368	0.444	0.404	0.393	0.425	0.425	0.398	0.361	0.341	0.311	
	taud	2.160	2.085	2.153	2.084	2.281	2.332	2.255	2.240	2.279	2.359	2.291	2.223	
	Ebn at Noon	687	761	818	793	852	864	826	804	782	745	650	623	
	Edh at Noon	76	107	121	146	126	121	128	124	107	82	68	62	
All-Sky Solar Radiation	RadAvg	0.93	1.86	3.10	4.24	5.71	6.07	5.81	5.14	3.39	1.88	0.80	0.57	
	RadStd	0.17	0.28	0.52	0.39	0.49	0.53	0.35	0.45	0.43	0.30	0.17	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.73	N/A	N/A	+0.89	N/A	N/A	-133	-203	+118	+59
(47) Regional (7 neighbors)		+0.77	N/A	N/A	+0.93	N/A	N/A	-156	-219	+122	+62

Nomenclature: See separate page